

PA3

0. PA3 gồm những gì?

Requirement 1 — Paper Prototype & Storyboarding

Chưa cần participant.

Team cần:

- 2 Main Scenarios
- 3 identified problems
- Mỗi problem có **3 alternative solutions**
- Tổng cộng **9 paper prototypes**
- Storyboard
- Explanation: Motivation, Problem, How it works, Strengths, Weaknesses
- Video demo

Requirement 2 — Informal/Formative Testing

Phần này mới cần participant.

Team cần:

- Testing objectives
- Target users
- Testing method
- Tasks/scenarios
- Criteria for success
- 2–3 participants
- Observations
- Feedback
- Unexpected behaviors/issues
- Points for improvement
- Compare alternatives
- Chọn prototype tốt nhất
- Improve prototype

Heuristic evaluation có thể dùng như kiểm tra nội bộ, nhưng **không thay thế participant testing**.

1. Bộ 9 prototypes

Problem	Prototype 1	Prototype 2	Prototype 3
P1 — Music Flow	1A Focus Run Mode	1B Adaptive Music	1C Ambient Audio Cue
P2 — Pace Awareness	2A Haptic Pace Feedback	2B Beat-to-Pace Matching	2C Just-in-Time Pace Alert
P3 — Interaction	3A Run Lock Interface	3B Earbud Controls	3C Voice-Free Quick Actions

Mỗi alternative phải là **một interaction idea khác nhau**, không phải chỉ đổi màu/UI.

2. Materials

- Giấy trắng

- Bút đen + bút màu
- Pencil/thước
- Sticky notes/giấy nhỏ
- Kéo + băng keo
- Điện thoại/smartwatch/earbuds thật nếu có
- Điện thoại/laptop để quay video

Không cần implement app thật.

3. Quy tắc paper prototype

3.1 Một trạng thái quan trọng = một sheet

Ví dụ:

```

RUNNING
↓
Pace becomes too fast
↓
Haptic feedback
↓
Runner slows down
↓
IN RANGE

```

3.2 Không cần vẽ đẹp

Mục tiêu là test:

- idea
- interaction
- workflow
- usability

3.3 Luôn ghi interaction

Ví dụ:

```

User taps PAUSE
↓
System shows confirmation

```

Hoặc:

```

User double-taps earbud
↓
Music pauses

```

4. Cách mô phỏng vibration

Không cần hardware. Dùng **Wizard-of-Oz**.

State	Pattern	Meaning
In range	No vibration	Keep pace
Too fast		Slow down
Too slow		Speed up

Trong video:

- Facilitator rung paper watch
- Hoặc dùng điện thoại thật đặt phía sau paper watch
- Hoặc mô tả bằng lời nếu cần

Khi testing, **không giải thích trước pattern có nghĩa gì.**

5. Cách mô phỏng audio

Không cần xây music system.

Có thể:

- dùng  MUSIC
- dùng  cue
- facilitator phát một âm thanh ngắn
- hoặc mô phỏng cue trong video

Mục tiêu là test concept, không phải chất lượng audio thật.

6. 1A — Focus Run Mode

Problem

Notification, calls và voice coaching có thể làm gián đoạn music.

Idea

Cho runner chọn những thứ cần block trước khi chạy.

Sheet 1

Running Music Coach

Ready to Run?

Target Pace: 6:00/km

 Running Mix

[START RUN]

Sheet 2

Focus Run

Block during run:

☒ Messages

☒ Calls

☒ Notifications

☒ Voice Coach

Emergency calls ☐

[CONFIRM]

Sheet 3

● RUNNING

```
6:02 / km

MUSIC
Music uninterrupted

[ PAUSE ]      [ END ]
```

Sheet 4

```
● RUNNING

Notification muted

Music continues

6:03 / km
```

Sheet 5

```
Run Complete

Distance    5.2 km
Time        31:20
Avg Pace    6:01/km

[ SAVE RUN ]
```

7. 1B — Adaptive Music

Problem

Music cố định có thể không phù hợp với trạng thái chạy.

Idea

Music thích ứng với pace mà không cần voice coaching.

Sheet 1

```
Choose Music

Running Mix

Music mode:
○ Original playlist
● Adaptive playlist

[ CONTINUE ]
```

Sheet 2

```
Adaptive Music

Target cadence
[ USER SETS ]

Music adapts to pace
● Gentle adaptation
```

○ Strong adaptation
[START]

Sheet 3

● RUNNING
6:02 / km
168 BPM
🎵 MUSIC
Adaptive mode: ON

Sheet 4

● RUNNING
5:40 / km
🎵 Music adapting...
No voice interruption

Sheet 5

● RUNNING
6:00 / km
✓ Pace stabilized
🎵 MUSIC

8. 1C — Ambient Audio Cue

Problem

Voice coaching có thể phá music flow.

Idea

Dùng audio cue ngắn thay vì lời nói dài.

Sheets

Sheet 1

● RUNNING
6:00 / km
🎵 MUSIC

Sheet 2

● RUNNING

5:35 / km
MUSIC
+ J cue

Sheet 3

● RUNNING
6:02 / km
MUSIC
No screen check

Sheet 4

● RUNNING
Pace deviation: 20 sec
J Cue
MUSIC

Sheet 5

● RUNNING
6:01 / km
✓ Stable
MUSIC

9. 2A — Haptic Pace Feedback

Problem

Runner không muốn nhìn màn hình để biết pace.

Idea

Vibration pattern.

Sheets

Sheet 1

● RUNNING
6:00/km
MUSIC
Keep pace

Sheet 2

5:30/km
T00 FAST

Sheet 3

6:05/km
✓
IN RANGE

Sheet 4

6:40/km
T00 SLOW

Sheet 5

6:02/km
✓
IN RANGE
MUSIC

10. 2B — Beat-to-Pace Matching

Problem

Runner có thể dùng music rhythm để hỗ trợ pace/cadence.

Idea

Match footsteps với beat.

Sheet 1

Pace Feedback
Target Pace: 6:00/km
Target Cadence: [SET]
[START]

Sheet 2

● RUNNING
MUSIC
Beat = cadence
6:00 / km

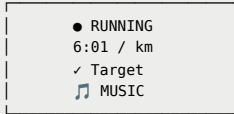
Sheet 3

● RUNNING
5:35 / km
Rhythm changes
Target beat shown

Sheet 4



Sheet 5



11. 2C — Just-in-Time Pace Alert

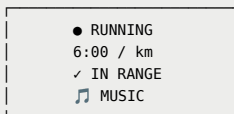
Problem

Alert quá thường xuyên sẽ interrupt runner.

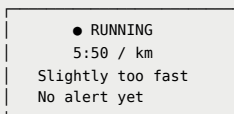
Idea

Chỉ alert khi deviation kéo dài.

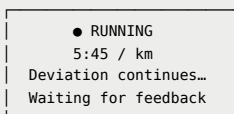
Sheet 1



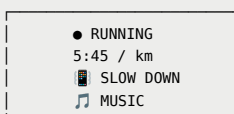
Sheet 2



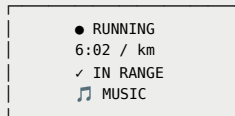
Sheet 3



Sheet 4



Sheet 5



12. 3A — Run Lock Interface

Problem

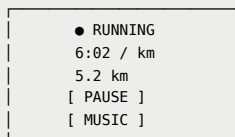
Touchscreen nhỏ + movement → accidental taps.

Idea

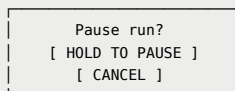
Minimal UI + large buttons + HOLD TO PAUSE.

Sheets

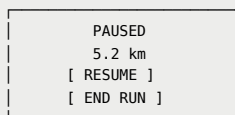
1



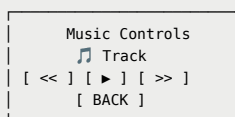
2



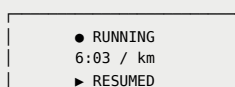
3



4



5



13. 3B — Earbud Controls

Problem

Runner không muốn lấy phone ra.

Idea

Earbud gestures.

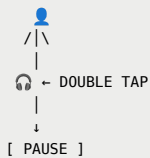
Sheet 1

Earbud Controls
Double tap → Pause
Swipe → Change track
Long press → Pace cue
[CONFIRM]

Sheet 2

● RUNNING
6:02 / km
MUSIC
Phone stays pocketed

Sheet 3



Sheet 4

● RUNNING
PAUSED
Earbuds
Phone untouched

Sheet 5

● RUNNING
▶ PLAYING
MUSIC
Phone stays pocketed

14. 3C — Voice-Free Quick Actions

Problem

Voice coaching/menu navigation làm gián đoạn runner.

Idea

2 quick actions:

- Double tap → Pause
- Long press → Pace cue

Sheet 1

Quick Actions
Double tap → Pause
Long press → Pace cue
[START]

Sheet 2

● RUNNING
6:02 / km
🎵 MUSIC
Quick actions ON

Sheet 3

👉
LONG PRESS
↓
Pace Cue
6:02 / km
✓ IN RANGE
🎵 MUSIC

Sheet 4

👉 👉
↓ ↓
● PAUSED
🎵 Music off
Double tap to resume

Sheet 5

● RUNNING
▶ RESUMED
🎵 MUSIC
6:03 / km

15. Storyboard

Mỗi prototype làm **4 frames**:

```
1. CONTEXT
  ↓
2. USER ACTION / TRIGGER
  ↓
3. SYSTEM RESPONSE
  ↓
4. OUTCOME
```

Mỗi frame nên có:

- Runner
- Environment
- Device
- Action
- Result
- 1 caption ngắn

Không cần vẽ UI chi tiết như paper screen.

16. Storyboard prompts

General style

Simple black ink hand-drawn UX storyboard, recreational runner, outdoor running environment, smartphone or smartwatch and earbuds, minimal white background, simple human figures, clear actions, no realistic details, no polished UI, academic UX storyboard style.

Focus Run

Frame 1:

A recreational runner preparing for an outdoor run, wearing earbuds and smartwatch, holding a smartphone. Simple black ink UX storyboard.

Caption: **Runner prepares to start a run.**

Frame 2:

Runner looking at a smartwatch and enabling a Focus Run mode before starting the run. Simple hand-drawn UX storyboard.

Caption: **Runner enables Focus Run.**

Frame 3:

Runner running outdoors while a phone notification is blocked or muted. Music continues through earbuds. Runner does not stop.

Caption: **Notifications are blocked while music continues.**

Frame 4:

Runner continues running comfortably without interacting with the phone.

Caption: **Runner continues without breaking music flow.**

Adaptive Music

Frame 1:

Runner selecting Adaptive Music on a smartwatch before an outdoor run.

Caption: **Runner starts with Adaptive Music.**

Frame 2:

| Runner running faster outdoors while smartwatch shows changing pace.

Caption: **Runner's pace changes.**

Frame 3:

| Runner continues running while music adapts, represented by musical notes changing around the earbuds.

Caption: **Music adapts without voice interruption.**

Frame 4:

| Runner maintains stable pace and continues listening to music without looking at the device.

Caption: **Runner maintains pace without breaking flow.**

Ambient Audio Cue

Frame 1:

| Runner listening to music while running outdoors, looking forward.

Caption: **Runner listens to music while running.**

Frame 2:

| Runner gradually runs too fast, smartwatch indicates pace deviation.

Caption: **Runner's pace becomes too fast.**

Frame 3:

| A subtle audio cue is represented by a small musical note near the runner's earbud.

Caption: **A subtle audio cue is provided.**

Frame 4:

| Runner adjusts pace and continues without checking the screen.

Caption: **Runner adjusts pace without looking at the screen.**

Haptic Pace Feedback

Frame 1:

| Runner running outdoors while wearing smartwatch and earbuds, looking forward.

Caption: **Runner continues running hands-free.**

Frame 2:

| Smartwatch produces two short vibration symbols while runner is running too fast.

Caption: **The watch gives two short vibration pulses.**

Frame 3:

| Runner slows down after feeling the vibration, without looking at the watch.

Caption: **Runner interprets the haptic feedback and slows down.**

Frame 4:

| Runner returns to target pace and continues running.

Caption: **Runner returns to the target pace.**

Beat-to-Pace Matching

Frame 1:

| Runner listening to music while running outdoors, musical notes around the earbuds.

Caption: **Runner runs with music rhythm.**

Frame 2:

| Runner's footsteps become faster than the target music rhythm.

Caption: **Runner's pace becomes too fast.**

Frame 3:

| Runner synchronizes footsteps with musical beat.

Caption: **Runner adjusts movement to match the rhythm.**

Frame 4:

| Runner maintains stable pace with synchronized footsteps and music.

Caption: **Runner maintains a stable rhythm and pace.**

Just-in-Time Pace Alert

Frame 1:

| Runner maintaining target pace while running outdoors.

Caption: **Runner is within the target pace.**

Frame 2:

| Runner becomes slightly too fast but receives no alert.

Caption: **Pace is slightly too fast, but no alert is triggered yet.**

Frame 3:

| Runner remains too fast and smartwatch provides a subtle alert.

Caption: **The deviation continues, so feedback is triggered.**

Frame 4:

| Runner slows down and returns to target pace.

Caption: **Runner corrects pace without constant interruptions.**

Run Lock

Frame 1:

| Runner looking at a minimal smartwatch running screen with large buttons.

Caption: **Runner views a simple run interface.**

Frame 2:

| Runner attempts to pause while moving.

Caption: **Runner attempts to pause the run.**

Frame 3:

| Smartwatch shows a large HOLD TO PAUSE interaction.

Caption: **System prevents accidental pause.**

Frame 4:

| Runner successfully pauses and resumes the run.

Caption: **Runner resumes with minimal interaction.**

Earbud Controls

Frame 1:

| Runner learns earbud gestures before starting the run.

Caption: **Runner learns the earbud gestures.**

Frame 2:

| Runner runs outdoors with phone inside pocket and earbuds in use.

Caption: **Phone stays in the pocket.**

Frame 3:

| Runner double-taps an earbud while running.

Caption: **Runner double-taps the earbud.**

Frame 4:

| Music pauses while phone remains untouched.

Caption: **Music pauses without touching the phone.**

Voice-Free Quick Actions

Frame 1:

| Runner learns double tap for pause and long press for pace cue.

Caption: **Runner learns the quick actions.**

Frame 2:

| Runner continues running outdoors while listening to music.

Caption: **Runner continues without opening the interface.**

Frame 3:

| Runner performs a long press gesture to request pace information.

Caption: **Runner long-presses for a quick pace cue.**

Frame 4:

| Runner double-taps to pause music without opening a menu.

Caption: **Runner controls the session without navigating menus.**

17. Explanation card

Mỗi prototype có 1 card:

PROTOTYPE:

PROBLEM:

What user problem are we solving?

MOTIVATION:

Why did we choose this idea?

HOW IT WORKS:

1.

2.
3.

STRENGTHS:

+
+
+

WEAKNESSES:

-
-
-

18. Video demo — Requirement 1

Video có thể ngắn.

Structure

1. Giới thiệu prototype
2. Nêu problem
3. Demo user workflow
4. Facilitator đổi paper sheets
5. Mô phỏng vibration/audio/gesture nếu có
6. Nêu solution

Ví dụ:

“This is Prototype 2A — Haptic Pace Feedback.”

“The problem is that runners need pace feedback without looking at the screen.”

Sau đó demo.

19. Requirement 2 — Participant Testing

Testing objectives

Có thể dùng:

1. Evaluate whether each prototype effectively addresses the identified user problem.
2. Evaluate usability and learnability of the proposed interaction.
3. Identify misunderstandings, errors and unexpected user behaviors.
4. Compare the three alternatives and identify the strongest direction.

20. Participants

2–3 participants.

Target:

- Recreational runners
- Có nghe music khi chạy
- Có dùng smartphone/smartwatch/earbuds khi chạy
- Không biết prototype trước khi test

21. Testing method

Dùng:

Moderated formative usability testing

Một người = Facilitator.

Một người = Participant.

Facilitator:

- đưa scenario/task
- không nói đáp án
- quan sát
- ghi behavior
- ghi feedback

22. Không nói đáp án trước

❌ Sai:

| “Bây giờ hãy double tap để pause.”

✅ Đúng:

| “You are running and want to pause your music. Show me what you would do.”

Nếu participant hỏi:

| “What should I press?”

Có thể trả lời:

| “What would you normally try?”

23. Metrics

Không cần quantitative experiment phức tạp.

Dùng 5 lightweight metrics:

Metric	Cách đo
Task completion	Success / Failure
Time-on-task	Thời gian hoàn thành task
Error count	Số thao tác sai
Assistance count	Số lần cần facilitator
Qualitative feedback	Participant nói gì

Haptic/audio: thêm Cue Interpretation Accuracy

Test:

- In range
- Too fast
- Too slow

Ví dụ:

Participant	Too fast	Too slow	In range
P01	✓	✓	✓
P02	✓	✗	✓

Participant	Too fast	Too slow	In range
P03	✓	✓	✓

24. Testing tasks

1A Focus Run

Scenario:

| You are about to start a solo run and want to avoid interruptions from notifications and calls.

Task:

| Prepare the run so that your music is not interrupted.

Observe:

- hiểu Focus Run?
- biết chọn block options?
- bỏ sót option?

1B Adaptive Music

Scenario:

| You are running and your pace changes during the session.

Task:

| Start a run and continue when your pace changes.

Observe:

- hiểu music adaptation?
- có cảm thấy mất control?

1C Ambient Audio

Scenario:

| You are running while listening to music. Your pace becomes too fast.

Task:

| Respond to the feedback without looking at the screen.

Observe:

- nhận ra cue?
- biết phải làm gì?

2A Haptic

Scenario:

| You are running and receive a vibration from your watch.

Task:

| Tell us what you think the vibration means and what you would do.

Không giải thích vibration trước.

Observe:

- đoán đúng?
- nhầm pattern?
- biết action tiếp theo?

2B Beat-to-Pace

Scenario:

| You are running faster than your target pace.

Task:

| Continue the run using the music as feedback.

Observe:

- hiểu rhythm?
- tự điều chỉnh?

2C Just-in-Time Alert

Scenario:

| You are slightly above your target pace.

Task:

| Continue running and respond when you receive feedback.

Observe:

- có nhận thấy system chưa alert?
- timing có hợp lý?
- hiểu alert?

3A Run Lock

Scenario:

| You want to pause your run.

Task:

| Pause the run.

Observe:

- tìm PAUSE?
- hiểu HOLD TO PAUSE?
- accidental action?

3B Earbud Controls

Scenario:

| You want to pause your music without taking your phone out.

Task:

| Pause the music.

Observe:

- discover gesture?

- nhớ gesture?

3C Quick Actions

Scenario:

| You want to quickly check your pace and then pause the music.

Task:

| Perform both actions.

Observe:

- discover long press?
- discover double tap?
- nhớ sequence?

25. Testing session

Step 1 — Briefing

Nói:

| "We are testing the prototype, not you. There are no right or wrong answers. Please tell us what you think while using it."

Step 2 — Background

1. How often do you run?
2. Do you listen to music while running?
3. What device do you normally use?
4. Do you normally check your pace during a run?

Step 3

Give scenario.

Step 4

Participant performs task.

Step 5

Ask:

- What did you think would happen?
- What did the feedback mean?
- What was confusing?
- What would you change?
- Would you use this while running? Why?

26. Observation sheet

Participant	Prototype	Task	Success	Time	Errors	Assistance	Observa
P01	2A	Interpret cue	✓	8s	0	0	Understc quickly

Participant	Prototype	Task	Success	Time	Errors	Assistance	Observa
P02	2A	Interpret cue	✗	12s	1	0	Confuse patterns
P03	2A	Interpret cue	✓	10s	0	0	Correct interpret

27. Feedback sheet

Prototype:

What did participant like?

What was confusing?

What did participant expect?

What would they change?

Would they use it?

YES / NO / MAYBE

Why?

28. Unexpected behavior

Đây là phần quan trọng.

Ví dụ:

- | Participant tried to swipe the watch instead of using the proposed button.
- | Participant interpreted two vibration pulses as confirmation rather than a warning.
- | Participant tried to control music from the smartwatch before discovering the earbud gesture.

Những hành vi này cho thấy mental model của user có thể khác với design.

29. Points of improvement

Dùng format:

ISSUE #1

Problem:

Observation:

Participant:

P01 / P02 / P03

Impact:

Possible improvement:

Ví dụ:

ISSUE #1

Problem:

Haptic patterns are difficult to distinguish.

Observation:

2/3 participants confused the vibration meaning.

Impact:

Runner may take the wrong action.

Possible improvement:

Make the two vibration patterns more distinct and test again.

30. Heuristic Evaluation

Không bắt buộc để hoàn thành Requirement 2.

Nếu muốn review nội bộ, có thể check:

- Visibility of system status
- Match between system and real world
- User control and freedom
- Consistency and standards
- Error prevention
- Recognition rather than recall
- Flexibility and efficiency
- Aesthetic/minimalist design
- Error recovery
- Help/documentation

Heuristic evaluation không thay participant testing.

Nếu cần severity:

0 = Not a usability problem
1 = Cosmetic
2 = Minor
3 = Major
4 = Critical

31. Compare 3 alternatives

Sau testing:

Criteria	Alternative A	Alternative B	Alternative C
Task success	3/3	2/3	3/3
Learnability	High	Medium	High
Errors	Low	Medium	Low
Assistance	Low	High	Low
Low interruption	High	High	Medium
User preference	High	Medium	Medium

Không bắt buộc phải cộng điểm. Có thể dùng qualitative comparison.

32. Chọn best prototype

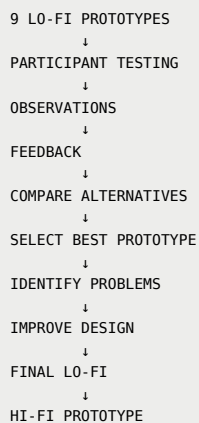
Ví dụ:

We selected Haptic Pace Feedback because all participants were able to complete the task with minimal assistance. However, two participants had difficulty distinguishing the vibration patterns. Therefore, the final design should use more distinguishable haptic patterns and test again.

Một kết luận tốt phải có:

- evidence
- weakness
- improvement

33. Flow cuối



34. Chia việc team

Member 1 — Problem 1

- 1A Focus Run
- 1B Adaptive Music
- 1C Ambient Audio
- Storyboards
- Video

Member 2 — Problem 2

- 2A Haptic
- 2B Beat-to-Pace
- 2C Just-in-Time
- Storyboards
- Video

Member 3 — Problem 3

- 3A Run Lock
- 3B Earbud
- 3C Quick Actions

- Storyboards
- Video

Sau khi xong cả team cùng:

- chuẩn hóa style
 - participant testing
 - tổng hợp results
 - chọn best prototypes
-

35. Checklist Requirement 1

Paper Prototype

- ☐ 1A Focus Run
- ☐ 1B Adaptive Music
- ☐ 1C Ambient Audio
- ☐ 2A Haptic
- ☐ 2B Beat-to-Pace
- ☐ 2C Just-in-Time Alert
- ☐ 3A Run Lock
- ☐ 3B Earbud
- ☐ 3C Quick Actions

Mỗi prototype

- ☐ Paper screens
 - ☐ Interaction annotations
 - ☐ Problem
 - ☐ Motivation
 - ☐ How it works
 - ☐ Strengths
 - ☐ Weaknesses
 - ☐ Storyboard
 - ☐ Video demo
-

36. Checklist Requirement 2

- ☐ Testing objectives
- ☐ Target user group
- ☐ Testing method
- ☐ Success criteria
- ☐ Test scenarios
- ☐ 2–3 participants
- ☐ Participants do not know prototype beforehand
- ☐ Observation notes
- ☐ Feedback

- ☐ Task completion
 - ☐ Time-on-task
 - ☐ Error count
 - ☐ Assistance count
 - ☐ Unexpected behavior
 - ☐ Issues
 - ☐ Points of improvement
 - ☐ Compare 3 alternatives
 - ☐ Select best prototype
 - ☐ Improve selected prototype
 - ☐ Final demo
-

37. Câu nhớ cho cả team

Requirement 1 = BUILD & EXPLAIN 9 IDEAS.

Requirement 2 = TEST → LEARN → COMPARE → IMPROVE → CHOOSE THE BEST.

Không cần làm prototype hoàn hảo ngay từ Requirement 1. Lo-fi prototype tồn tại để team phát hiện vấn đề càng sớm càng tốt trước khi làm hi-fi.